# HW6 Solutions

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1. The Jacobson algorithm updates the Estimated “RTT” using the formula: $\mathrm{RTT}_{\mathrm{new}} = \alpha \cdot \mathrm{RTT}_{\mathrm{old}} + (1 - \alpha) \cdot \mathrm{RTT}_{\mathrm{sample}}$ Given $\alpha = 0.9$ and initial $\mathrm{RTT} = 30$ msec:
   - After 26 msec: $\mathrm{RTT} = 0.9 \cdot 30 + 0.1 \cdot 26 = 27 + 2.6 = 29.6$ msec.
   - After 32 msec: $\mathrm{RTT} = 0.9 \cdot 29.6 + 0.1 \cdot 32 = 26.64 + 3.2 = 29.84$ msec.
   - After 24 msec: $\mathrm{RTT} = 0.9 \cdot 29.84 + 0.1 \cdot 24 = 26.856 + 2.4 = 29.256$ msec.

   The new RTT estimate is 29.256 msec.

2. We start with cwnd = 2 KB (1 MSS) and rwnd = 24 KB. The RTT is 10 msec.
   - $t = 0$ ms: Send 1 MSS (2 KB).
   - $t = 10$ ms: Receive ACK, cwnd becomes 2 MSS (4 KB). Send 2 MSS.
   - $t = 20$ ms: Receive ACKs, cwnd becomes 4 MSS (8 KB). Send 4 MSS.
   - $t = 30$ ms: Receive ACKs, cwnd becomes 8 MSS (16 KB). Send 8 MSS.
   - $t = 40$ ms: Receive ACKs, cwnd becomes 16 MSS (32 KB). However, the sender is limited by rwnd = 24 KB (12 MSS).

   It takes 40 msec before the first full window (24 KB) can be sent.

3. In TCP Tahoe, when a timeout occurs:
   - ssthresh is set to half of the current cwnd: $\frac{128}{2} = 64$ KB.
   - cwnd is reset to 1 MSS (1 KB).
   - Slow start occurs until cwnd reaches ssthresh, then congestion avoidance begins.
   - Bursts 1-6 (Slow Start): cwnd doubles each time ($1, 2, 4, 8, 16, 32 \to 64$).
   - Bursts 7-10 (Congestion Avoidance): cwnd increases by 1 MSS per burst.
     - Burst 7: 65 KB
     - Burst 8: 66 KB
     - Burst 9: 67 KB
     - Burst 10: 68 KB

   The window will be 68 KB.

4. cwnd = 18 KB, Timeout occurs. MSS = 1 KB.
   - ssthresh = $\frac{18}{2} = 9$ KB.
   - cwnd = 1 KB.
   - Burst 1: cwnd becomes 2 KB.
   - Burst 2: cwnd becomes 4 KB.
   - Burst 3: cwnd becomes 8 KB.
   - Burst 4: cwnd reaches ssthresh = 9 KB.

The window will be 9 KB.

5. Given: Window = 65,535 bytes, Capacity = 1 Gbps, One-way delay = 10 ms (RTT = 20 ms).
   - Maximum Throughput = $\frac{\text{Window Size}}{\text{RTT}}$ $\text{Throughput} = \frac{65535 \cdot 8 \text{ bits}}{0.020 \text{ sec}} = 26,214,000 \text{ bps} = 26.2 \text{ Mbps}$.
   - Line Efficiency = $\frac{\text{Throughput}}{\text{Capacity}}$ $\text{Efficiency} = \frac{26.214 \text{ Mbps}}{1000 \text{ Mbps}} = 0.026214$.

   Maximum throughput: 26.2 Mbps

   Line efficiency: 2.6%

6. With an 8-bit sequence number, there are $2^8 = 256$ unique sequence numbers. To avoid wrap-around within the maximum TPDU lifetime ($T = 30$ sec):
   - Total data = $256 \cdot 128 \text{ bytes} = 32,768 \text{ bytes}$.
   - Max Data Rate = $\frac{32,768 \cdot 8 \text{ bits}}{30} \text{ sec} \approx 8,738.1 \text{ bps}$.

   The maximum data rate is approximately 8.7 kbps.

7. The maximum length of an IP packet is 65,535 bytes (defined by a 16-bit length field). A standard IP header is 20 bytes and a standard TCP header is 20 bytes. $65,535 - 20 - 20 = 65,495$ bytes. This number was chosen to allow a TCP segment to fit into the largest possible IP packet without fragmentation.

8. The sender's transmission is limited by the minimum of the congestion window and the receiver's advertised window: $\text{Allowed} = \min(\text{rwnd}, \text{cwnd}) = \min(100 \text{ KB}, 50 \text{ KB}) = 50 \text{ KB}$. The sender can transmit 50 KB.

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